E-Commerce Data Analytics Platform

Project 02 — Business Development & Technical Overview

GitHub Portfolio | Data Analytics | SQL · Python · Power BI

1. Executive Summary

This project delivers a full-stack e-commerce data analytics pipeline — from raw API ingestion through to interactive Power BI dashboards. It demonstrates the ability to design, build, and operationalize a modern data workflow that transforms unstructured product data into actionable business intelligence.

The system integrates three key components: a Python-based ETL pipeline consuming the FakeStore REST API, a SQL Server data warehouse supporting complex analytical queries, and a Power BI reporting layer. A supplementary Superstore Orders dataset enriches the solution with retail sales, profitability, and shipping analytics.

2. Business Context & Problem Statement

E-commerce businesses generate vast volumes of transactional and product data daily. Without a structured analytics layer, leadership cannot confidently answer critical questions such as:

- Which product categories generate the highest revenue and margin?
- What is the price sensitivity across customer segments?
- Which products are trending based on ratings and review volume?
- How efficiently are orders being shipped, and where are delays concentrated?
- What discount strategies maximize profit without eroding margins?

This project solves these challenges by building a reusable, scalable analytics platform that automates data collection, cleans and enriches raw records, and surfaces insights through a governed reporting layer.

3. Solution Architecture

3.1 Technology Stack

Layer	Technology	Purpose
Data Source	FakeStore API (REST)	Real-time product data ingestion

Layer	Technology	Purpose
Processing	Python · Pandas	Data cleaning, transformation, feature engineering
Storage	SQL Server (ecommerce_db)	Relational data warehouse for analytical queries
Analytics	SQL Server · T-SQL	Advanced querying, aggregations, KPI derivation
Visualization	Power BI (.pbix)	Interactive dashboards and business reporting

3.2 Data Flow

- FakeStore API is called via Python requests to retrieve live product catalogue data (20 records across 4 categories).
- Pandas performs data cleaning: rating objects are flattened, prices rounded, text fields normalised, and category labels standardised.
- Feature engineering adds Price Category (Low / Medium / High), Popularity Score (rating × review count), and Title Length for NLP-readiness.
- Cleaned data is exported to CSV and bulk-loaded into SQL Server via pyodbc for persistence.
- T-SQL queries enable advanced analytics: GROUP BY aggregations, window functions, date arithmetic, CTEs, and subqueries.
- Power BI connects to SQL Server to render dashboards with drill-through capability and dynamic filtering.

4. Key Business Metrics & KPIs

KPI / Metric	Description	Business Value
Avg. Price by Category	Mean product price per category	Informs pricing and competition strategy
Popularity Score	Rating × Review Count	Identifies top-selling & trending products
Price Tier Distribution	Low / Medium / High segments	Guides promotional and discount planning
Total & Profit Analysis	Aggregated sales, profit, discount (Superstore)	P&L visibility, cost control, margin tracking
Shipping Lead Times	Business-day difference order → ship	Operational efficiency and SLA monitoring

5. Core Analytical Capabilities

5.1 SQL Analytics (Superstore & E-Commerce)

- Advanced filtering with WHERE, IN, BETWEEN, LIKE, and NULL handling.
- Aggregation functions: SUM, COUNT, AVG, MIN, MAX with GROUP BY and HAVING.
- Window-style analytics using subqueries and CTEs to compute rolling metrics.
- Business-day shipping calculation using DATEDIFF with week-offset correction.
- Year-over-year sales comparison using conditional aggregation (CASE WHEN + DATEPART).
- String manipulation: REPLACE, TRANSLATE, CHARINDEX, CONCAT, UPPER for data quality.
- Multi-table JOIN patterns: INNER, LEFT JOIN, and self-joins for hierarchical relationships.
- ICC World Cup win/loss analysis demonstrating UNION ALL + subquery pattern.

5.2 Python Data Engineering

- REST API consumption and JSON normalisation with Pandas DataFrame.
- Lambda functions for nested object extraction (rating.rate, rating.count).
- Column naming standardisation via custom camelCase transformer.
- Business rule encoding with price_category() classification function.
- Computed feature: Popularity_Score = rating_rate × rating_count.
- End-to-end automation: API → clean CSV → SQL Server insert in a single notebook run.

5.3 Power BI Visualisation

- Interactive dashboard built from the SQL Server ecommerce_db and Superstore datasets.
- Filters and slicers across category, region, segment, ship mode, and date range.
- KPI cards, bar/column charts, and trend lines for executive-level reporting.

6. Business Value & Impact

This platform delivers tangible value across multiple business functions:

Merchandising & Pricing:

- Automated price-tier segmentation enables targeted promotions without manual classification.
- Category-level margin analysis reveals which product lines to scale or deprioritise.

Operations & Logistics:

- Business-day shipping analysis identifies fulfilment bottlenecks by city and region.
- Discount-to-profit ratio reporting prevents margin erosion from over-discounting.

Customer & Growth:

- Popularity Score surfaces viral products for featured placement and ad investment.
- Segment-based analysis (Consumer, Corporate, Home Office) enables personalised marketing.

7. Project Deliverables

- Project_02.sql — Full T-SQL script covering DML, DDL, joins, CTEs, date and string functions.
- Project_02_DA.ipynb — Python Jupyter Notebook for API ingestion, data cleaning, and SQL Server loading.
- project_002.pbix — Power BI dashboard file with interactive visualisations connected to the data warehouse.
- cleaned_products.csv — Exported, analysis-ready dataset from the Python pipeline.

8. Skills Demonstrated

This project is designed to showcase a well-rounded, industry-relevant data analytics skill set suitable for roles in Data Analysis, Business Intelligence, and Data Engineering.

- SQL (T-SQL): Intermediate to Advanced — DDL/DML, window functions, CTEs, subqueries, date arithmetic.
- Python: Data Engineering — API consumption, Pandas ETL, feature engineering, database connectivity.
- Power BI: Dashboard design — KPIs, slicers, multi-page reports, SQL Server integration.
- Data Architecture: End-to-end pipeline design from source to insight.
- Business Acumen: KPI definition, pricing strategy, and operational analytics.

9. How to Run This Project

Prerequisites:

- Python 3.9+ with pandas, requests, pyodbc installed.
- SQL Server (local or remote) with ecommerce_db and Superstore_orders databases created.
- Power BI Desktop to open the .pbix report file.

Steps:

- Run Project_02.sql in SQL Server Management Studio to set up schema and seed Superstore data.
- Execute Project_02_DA.ipynb cell-by-cell to pull FakeStore API data, clean it, and load into SQL Server.
- Open project_002.pbix in Power BI Desktop and refresh the data source connection.

— End of Document —